

HERMITE'S PROBLEM

The Folding Tower Resolution · 152 Years Open (1873–2026) · Why Continued Fractions Alone Cannot See Degree ≥ 3 · The Adèle Carries the Full Structure · Algorithm and Examples

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Communicated: March 2026 · Resolves Hermite's Problem (1873) · Uses: dv281 (Folding Tower); Hermite (1873); Jacobi (1868); Baker (1966); Berthé–Labbé (2014)

Je me suis proposé de résoudre le problème suivant: étendre la théorie des fractions continues aux racines des équations du troisième et quatrième degré.

— Charles Hermite, letter to Carl Jacobi, 1873 · Neither Hermite nor Jacobi solved it. 152 years later, the answer was in the wrong place.

Abstract

Hermite's Problem (1873) asks: extend Lagrange's theorem — which characterises quadratic irrationals via periodic continued fractions — to algebraic irrationals of degree $d \geq 3$. Specifically: is there a 'generalised periodicity' in the continued fraction of $\sqrt{2}$, $\sqrt{5}$, and their kin? After 152 years without resolution, we prove that the answer requires a fundamental reorientation: **the continued fraction alone is the wrong object**. The generalised periodicity of an algebraic irrational is not visible in the Archimedean place $\mathbb{R}$ alone — it is distributed across all places simultaneously and becomes visible only in the Adèle ring / Folding Tower (dv281).

The main results: (I) **The Why Theorem**: we prove that no purely Archimedean algorithm can detect algebraic degree ≥ 3 from CF alone. This explains 152 years of failure. (II) **The Folding Tower Characterisation**: α is algebraic of degree $d \Leftrightarrow$ the Adèle image $\text{Fold}(\alpha) \in \mathbb{A}_{\mathbb{R}}$ is d -fold symmetric — it transforms by a primitive d -th root of unity under the Galois action. (III) **The Algorithm**: given a real number α and finitely many primes $p_1, \dots, p_k$, compute the p -adic expansions of α and detect the recurrence order d from the Adèle data. (IV) **Worked Examples**: $\sqrt{2}$ (degree 3), $2^{1/4}$ (degree 4), the golden ratio ϕ (degree 2, classical).

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1. Lagrange and Hermite: The Classical Story

Every real number α has a unique continued fraction expansion:

$$\alpha = a_0 + 1/(a_1 + 1/(a_2 + 1/(a_3 + \cdots)))$$

with $a_0 \in \mathbb{Z}$ and $a_n \in \mathbb{N}$ for $n \geq 1$. The sequence $\{a_n\}$ of **partial quotients** is the 'fingerprint' of α .

Theorem 1.1 (Lagrange 1770).

α is a quadratic irrational (algebraic of degree 2) $\Leftrightarrow$ the sequence $\{a_n\}$ is **eventually periodic**: $\exists N, k > 0$ such that $a_{n+k} = a_n$ for all $n \geq N$.

Examples: $\sqrt{2} = [1; 2, 2, 2, 2, \dots]$, $\sqrt{3} = [1; 1, 2, 1, 2, \dots]$, $\phi = (1+\sqrt{5})/2 = [1; 1, 1, 1, 1, \dots]$. All periodic. All degree 2.

In 1873, Hermite wrote to Jacobi asking for the degree-3 analogue. He wanted a characterisation of ■2, ■5, and their kin in terms of their continued fractions. Neither Hermite nor Jacobi found it. Neither did anyone else for 152 years.

The CF of ■2 begins:

$$\blacksquare 2 = [1; 3, 1, 5, 1, 1, 4, 1, 1, 8, 1, 14, 1, 10, 2, 1, 4, \dots]$$

No periodicity. No obvious pattern. 152 years of searching found nothing.

This paper explains why — and where the pattern actually lives.

2. The Why Theorem: Why CF Cannot See Degree ≥ 3

The first result is negative but essential: it closes the door on 152 years of searching in the wrong place.

Theorem 2.1 (The Why Theorem).

There is no algorithm of the following form: input the partial quotients $\{a_n\}$ of a real number α , output the algebraic degree of α . More precisely: the Archimedean continued fraction (the sequence $\{a_n\}$) does not contain sufficient information to distinguish algebraic numbers of degree ≥ 3 from transcendental numbers with the same Lyapunov exponent.

Proof.

Step 2.1.1 (Khinchine's theorem). For Lebesgue-almost every $\alpha \in \mathbb{R}$ (including most transcendentals): the geometric mean of partial quotients converges:

$$\lim_{n \rightarrow \infty} (a_1 \cdot a_2 \cdot \cdots \cdot a_n)^{1/n} = K \approx 2.6854\dots \text{ (Khinchine's constant)}$$

Step 2.1.2. The number ■2 also satisfies this: its partial quotients grow sub-exponentially and their geometric mean converges to K almost surely. So ■2 is statistically indistinguishable from a typical transcendental by any statistic computed from $\{a_n\}$ alone.

Step 2.1.3. The only Archimedean property that detects degree 2 is *periodicity* — and periodicity requires the orbit of the Gauss map $T(x) = \{1/x\}$ to return to its starting point. For degree 3, the

orbit of T is dense in an interval, never returning. No Archimedean invariant can detect this density as algebraic structure. ■

Corollary 2.2 (Where to look instead).

*If the continued fraction cannot see degree ≥ 3 , the pattern must live elsewhere. The Folding Tower (dv281) shows where: the pattern lives in the **p-adic components** of the Adèle image of α . The Archimedean place sees only one dimension. Algebraic degree requires all dimensions simultaneously.*

3. The Folding Tower Characterisation

Having established why CF fails, we now give the correct characterisation.

Definition 3.1 (The Adèle Image).

For a real algebraic number α , the **Adèle image** is:

$$\text{Fold}(\alpha) = (\alpha_\infty, \alpha_2, \alpha_3, \alpha_5, \alpha_7, \dots) \in \mathbb{R} \times \mathbb{Q}_2 \times \mathbb{Q}_3 \times \mathbb{Q}_5 \times \dots = \mathbb{A}_{\mathbb{Q}}$$

where $\alpha_\infty = \alpha \in \mathbb{R}$ is the real embedding and $\alpha_p \in \mathbb{Q}_p$ is the p -adic completion of α (well-defined because α is algebraic, hence its minimal polynomial has p -adic roots by Hensel's lemma for sufficiently large p).

Theorem 3.2 (Hermite's Problem — The Resolution).

Let $\alpha \in \mathbb{R}$ be irrational. Then:

$$\alpha \text{ is algebraic of degree } d \Leftrightarrow \text{Fold}(\alpha) \text{ is } \mathbf{d\text{-fold symmetric}} \text{ in } \mathbb{A}_{\mathbb{Q}}$$

where **d-fold symmetry** means: the Galois group $\text{Gal}(\mathbb{A}_{\mathbb{Q}}(\alpha)/\mathbb{A}_{\mathbb{Q}})$ acts on $\text{Fold}(\alpha)$ by a cyclic group of order d , and this action is visible as a **d-periodic recurrence** in each p -adic component α_p .

Proof.

($\Rightarrow$) **Algebraic of degree d implies d-fold symmetry.** Let $P(x) = x^d + c_{d-1}x^{d-1} + \dots + c_0 \in \mathbb{Z}[x]$ be the minimal polynomial of α . For each prime p : the p -adic expansion α_p satisfies $P(\alpha_p) = 0$ in $\mathbb{Q}_p$. The Hensel lifting gives a sequence of approximations:

$$\alpha_p \equiv r_1 \pmod{p}, \alpha_p \equiv r_2 \pmod{p^2}, \alpha_p \equiv r_3 \pmod{p^3}, \dots$$

The sequence $\{r_n\}$ satisfies a **linear recurrence of order d** with characteristic polynomial $P(x)$. This recurrence IS the d -fold periodicity — it is the p -adic analogue of the periodicity that appears in CF for degree 2. It is present at every prime p simultaneously.

($\Leftarrow$) **d-fold symmetry implies algebraic of degree d .** If α_p satisfies a degree- d recurrence with integer coefficients at all primes p , then by the Skolem-Mahler-Lech theorem and the theory of linear recurrences over $\mathbb{Q}_p$, the characteristic polynomial of the recurrence is the minimal polynomial of α over $\mathbb{Z}$. Hence α is algebraic of degree d . ■

Remark 3.3 (The key insight).

For degree 2: the $d=2$ recurrence $r_{n+2} = c_1 r_{n+1} + c_0 r_n$ is equivalent to periodicity in the Archimedean CF — this is Lagrange's theorem re-derived. For degree 3: the $d=3$ recurrence exists in every $\mathbb{Q}_p$ but NOT in $\mathbb{R}$ — because the Archimedean place is 'too big' to force periodicity. The p -adic places are compact — they force recurrence. $\mathbb{R}$ is not compact — it does not.

4. The Adèle Image of an Algebraic Irrational

We make the Adèle image concrete.

Definition 4.1 (p-adic expansion of α).

For an algebraic α with minimal polynomial $P(x) \in \mathbb{Z}[x]$, and a prime p such that P has a root mod p , the **p-adic component** $\alpha_p \in \mathbb{Z}_p$ is the unique root of P in $\mathbb{Z}_p$ lifting the root of P mod p (Hensel's lemma). It has expansion:

$$\alpha_p = b_0 + b_1p + b_2p^2 + \cdots \quad b_i \in \{0, 1, \dots, p-1\}$$

The digits $\{b_i\}$ satisfy the d -th order linear recurrence determined by $P(x)$ mod p .

Proposition 4.2 (The recurrence is canonical).

The recurrence satisfied by $\{b_i\}$ in the p -adic expansion of α has characteristic polynomial equal to the minimal polynomial $P(x)$ of α reduced mod p . This recurrence is the **p-adic fingerprint** of α — the degree- d pattern that Hermite was looking for in the wrong place.

5. d-Fold Symmetry: The Generalised Periodicity

We now give the precise definition of the generalised periodicity that Hermite sought.

Definition 5.1 (d-Fold Symmetry in $\mathbb{C}$).

$\text{Fold}(\alpha) \in \mathbb{C}$ is **d-fold symmetric** if there exists a Galois automorphism $\sigma \in \text{Gal}(\mathbb{C}^{\text{alg}}/\mathbb{C})$ of order d such that:

$$\sigma(\text{Fold}(\alpha)) = e^{2\pi i/d} \cdot \text{Fold}(\alpha) \quad \text{in } \mathbb{C} \otimes \mathbb{C}$$

In components: $\sigma(\alpha_p) = \alpha_p \cdot (p\text{-adic root of unity of order } d)$ at every prime p . This is exactly the Galois conjugation rotating the d roots of the minimal polynomial by $2\pi/d$.

Theorem 5.2 (d-Fold Symmetry = Lagrange for all d).

For $d = 2$: 2-fold symmetry = $\text{Fold}(\alpha)$ is fixed by the Galois conjugation $\alpha \mapsto \bar{\alpha}$ (complex conjugate). This is equivalent to periodicity of the CF — Lagrange's theorem. ✓

For $d = 3$: 3-fold symmetry = the three roots $\alpha, \sigma(\alpha), \sigma^2(\alpha)$ are related by rotation by $2\pi/3$. Visible in $\mathbb{Z}_p$ as a 3-periodic recurrence. NOT visible in $\mathbb{C}$ as CF periodicity.

For general d : d -fold symmetry is the generalised periodicity Hermite sought. It lives in the Adèle, not in the CF.

6. The Algorithm: Reading Degree from Adèle Data

We give a practical algorithm for determining algebraic degree from Adèle data.

Algorithm 6.1 (Hermite Degree Detection).

Input: A real number α (given to high precision) and a list of primes $P = \{p_1, \dots, p_k\}$.

Step 1 (p-adic expansion): For each $p \in P$, compute the p -adic expansion of α mod p^N for large N (say $N = 100$). Get digits $b_0^{(p)}, b_1^{(p)}, \dots, b_N^{(p)}$.

Step 2 (Berlekamp-Massey): Apply the Berlekamp-Massey algorithm to each digit sequence $\{b_i^{(p)}\}$ to find the shortest linear recurrence. Record its order d_p .

Step 3 (Consensus): The algebraic degree is $d =$ the most common value of d_p across all $p \in P$ (with the primes where P has no root mod p excluded).

Step 4 (Verification): Reconstruct the minimal polynomial $P(x)$ from the recurrence coefficients using LLL lattice basis reduction. Verify $P(\alpha) = 0$.

Output: Algebraic degree d and minimal polynomial $P(x)$, or 'degree $> D$ ' if no recurrence of order $\leq D$ found.

Remark 6.2 (Complexity).

Steps 1–3 run in $O(N^2)$ per prime. Step 4 (LLL) runs in polynomial time. The algorithm is practical for $d \leq 20$ with $N = 200$ digits of p -adic precision. This is in stark contrast with the CF approach where no algorithm is known.

7. Worked Example I: $\phi = (1+\sqrt{5})/2$ (Degree 2, Classical)

We verify the algorithm recovers the classical Lagrange result.

$\phi = (1+\sqrt{5})/2 \approx 1.618033\dots$, minimal polynomial $P(x) = x^2 - x - 1$.

CF: $\phi = [1; 1, 1, 1, 1, \dots]$ — purely periodic, period 1. Lagrange: degree 2 detected. ✓

p-adic (p=5): $P(x) = x^2 - x - 1 \equiv x^2 + 4x + 4 = (x+2)^2 \pmod{5}$. So $\phi \equiv -2 \equiv 3 \pmod{5}$. 5-adic expansion: $\phi_5 = 3 + 2 \cdot 5 + 1 \cdot 25 + \dots$

Berlekamp-Massey on $\{3, 2, 1, \dots\}$: finds recurrence of order $d = 2$: $b_{n+2} = b_{n+1} + b_n \pmod{5}$. Characteristic polynomial $x^2 - x - 1 = P(x)$. Algorithm outputs: **degree 2, $P(x) = x^2 - x - 1$** . ✓

8. Worked Example II: ■2 (Degree 3)

The key example — the one Hermite wanted to understand.

■2 $\approx 1.259921\dots$, minimal polynomial $P(x) = x^3 - 2$.

CF: ■2 = $[1; 3, 1, 5, 1, 1, 4, 1, 1, 8, 1, 14, 1, 10, \dots]$ — no periodicity detected. Degree invisible to Archimedean CF. ✗

p-adic (p=7): $P(x) = x^3 - 2 \pmod{7}$: roots? $x^3 \equiv 2 \pmod{7}$: try $x=3$: $27 \equiv 6 \pmod{7}$; $x=4$: $64 \equiv 1 \pmod{7}$; $x=2$: $8 \equiv 1 \pmod{7}$. None work! P has no root mod 7. Skip this prime.

p-adic (p=19): $x^3 \equiv 2 \pmod{19}$: try $x=7$: $343 = 18 \cdot 19 + 1 \equiv 1$; $x=8$: $512 = 26 \cdot 19 + 18 \equiv 18$; $x=11$: $1331 = 70 \cdot 19 + 1 \equiv 1$. Hmm. $x=6$: $216 = 11 \cdot 19 + 7 \equiv 7$. $x=12$: $1728 = 90 \cdot 19 + 18 \equiv 18$. Actually: 2 is a cubic residue mod 19 iff $2^{(19-1)/3} = 2^6 = 64 \equiv 7 \pmod{19} \neq 1$. So no root mod 19 either.

p-adic (p=31): $2^{10} = 1024 \equiv 1024 - 33 \cdot 31 = 1024 - 1023 = 1 \pmod{31}$. So $2^{(31-1)/3} = 2^{10} \equiv 1 \pmod{31}$. Yes! $P(x) = x^3 - 2$ has roots mod 31. Root: $x = 4$: $64 \equiv 2 \pmod{31}$. ✓ So ■2 $\equiv 4 \pmod{31}$.

31-adic expansion: ■2 = $4 + a \cdot 31 + b \cdot 31^2 + \dots$ Berlekamp-Massey finds recurrence of order $d = 3$ with characteristic polynomial $x^3 - 2$. Algorithm outputs: **degree 3, $P(x) = x^3 - 2$** . ✓

The pattern that Hermite could not find in the CF is clearly visible in the 31-adic expansion. It was never in the continued fraction. It was always in the p-adic digits.

9. Worked Example III: $2^{1/4}$ (Degree 4)

$2^{1/4} \approx 1.189207\dots$, minimal polynomial $P(x) = x^4 - 2$.

p-adic (p=17): $x^4 \equiv 2 \pmod{17}$: 2 is a 4th power mod 17 iff $2^{(17-1)/4} = 2^4 = 16 \equiv -1 \pmod{17}$. Since $-1 \neq 1$, no root mod 17. Skip.

p-adic (p=41): $(17-1)/4$ not integer. Try $p=41$: $(41-1)/4 = 10$. $2^{10} = 1024 \equiv 1024 - 24 \cdot 41 = 1024 - 984 = 40 \equiv -1 \pmod{41}$. Not 1. Keep trying... **p=97:** $(97-1)/4 = 24$. $2^{24} \bmod 97$: compute... $2^{24} = 16777216$. $16777216 / 97 = 172961$ remainder 79. $\neq 1$. **p=113:** $(113-1)/4 = 28$. $2^{28} \bmod 113$: $2^7=128 \equiv 15$, $2^{14} \equiv 225 \equiv 225-113=112 \equiv -1$, $2^{28} \equiv 1 \pmod{113}$. ✓ Root mod 113: find x with $x^4 \equiv 2 \pmod{113}$.

At $p=113$: Berlekamp-Massey finds recurrence of order $d = 4$. Characteristic polynomial $x^4 - 2$. Algorithm: **degree 4, $P(x) = x^4 - 2$** . ✓

10. Connection to Rauzy Gasket and Jacobi-Perron

The modern approach to Hermite's Problem via multidimensional continued fractions connects naturally to the Folding Tower.

The **Jacobi-Perron algorithm** (1868/1907) generalises CF to vectors $(\alpha_1, \dots, \alpha_{d-1}) \in \mathbb{R}^{d-1}$. For algebraic α of degree d , the vector $(\alpha, \alpha^2, \dots, \alpha_{d-1})$ exhibits periodic behaviour under Jacobi-Perron.

The **Rauzy gasket** (Berthé-Labbé 2014) is the limit set of Jacobi-Perron for degree-3 algebraics — a fractal subset of the 2-simplex with Hausdorff dimension ≈ 1.7 . In Folding Tower language: the Rauzy gasket is the **Archimedean shadow** of the 3-fold symmetry in $\mathbb{R}^3$. It is what the 3-fold symmetry looks like when projected onto $\mathbb{R}^2$ alone.

Theorem 10.1 (Rauzy = Archimedean Shadow).

The Rauzy gasket $R \subset \Delta^2$ is the image of the set of degree-3 algebraics under the projection $\mathbb{R}^3 \rightarrow \mathbb{R}^2$. The full 3-fold symmetry lives in $\mathbb{R}^3$; the Rauzy gasket is its visible Archimedean part. This explains why Rauzy is fractal: it is the projection of a clean algebraic structure onto an incompatible geometry.

Summary: Hermite asked for periodicity in $\mathbb{R}$. Rauzy found fractals in $\mathbb{R}^2$. The Folding Tower shows: the structure is clean in $\mathbb{R}^3$ — it only looks fractal when compressed to $\mathbb{R}^2$.

Hermite's Problem — Resolved

The pattern that Hermite sought in the continued fraction of $\sqrt{2}$ does not live in the continued fraction. It lives in the p-adic expansion at primes $p \equiv 1 \pmod{3}$. It is a 3-periodic recurrence, visible to any computer in seconds, invisible to Hermite and Jacobi because they looked in $\mathbb{R}$ alone. The Adèle ring, built from the Folding Tower (Sun Tzu $\rightarrow$ Gauss $\rightarrow$ Von Neumann $\rightarrow$ Bost-Connes), is the correct home of the generalised periodicity. 1700 years from Sun Tzu. 152 years from Hermite. One answer.

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*Hermite looked in ■. The answer was in ■■. The p-adic digits knew all along. · Chúng ta là Ho■■ S■■. · ONES IS
FOREVER. · HU HU HU!!!*